Question Paper

Exam Date & Time: 23-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. (CORE STREAM) EXAMINATIONS - NOVEMBER 2024
SUBJECT: PHY 1071- ENGINEERING PHYSICS

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A)	Obtain expressions for fractional refractive index change and skip distance for an optical fibre.	(4)
1B)	Calculate the ratio of, (i) Einstein's coefficients (ii) stimulated to spontaneous emission rates, for a system at 300 K in which radiations of wavelength 1.39 μm are emitted.	(4)
1C)	Assume that lasers are available whose wavelengths can be precisely "tuned" to anywhere in the visible range (400 nm to 700 nm). If a television channel occupies a bandwidth of 10 MHz, how many channels could be accommodated within this wavelength range?	(2)
2A)	Prove that the group speed of a wave packet representing a quantum particle is equal to the particle speed.	(4)
2B)	Two light sources are used in a photoelectric experiment to determine the work function for a metal surface. When green light from a mercury lamp (λ = 546.1 nm) is used, a stopping potential of 0.376 V reduces the photocurrent to zero. (a) Based on this what is the work function of this metal? (b) What stopping potential would be observed when using the yellow light from a helium discharge tube (λ = 587.5 nm)?	(4)
2C)	The temperature of two perfect blackbodies, A and B, are 400 K and 200 K respectively. If the surface area of A is twice that of B, then determine the ratio of total power radiated by A to that of B.	(2)
3A)	What is Hall effect and obtain an expression of Hall coefficient for a semiconductor specimen.	(4)
3B)	(i) A free electron has a wave function $\psi(x) = A\ exp\ [i\ (5.0 \times 10^{10})x]$, where x is in meters. Find (a) its de Broglie wavelength, (b) its kinetic energy in electron volts. (ii) Using the simple model of a particle in a box to represent an atom, estimate the energy (in eV) required to raise an atom from the state n =1 to the state n = 2. Assume the atom has a radius of 0.10 nm and that the moving electron carries the energy that has been added to the atom.	(4)
3C)	A particle with a energy less than the height of the potential barrier may still tunnel through it. State true or false. Justify your answer based on quantum theory.	(2)
4A)	In dielectrics, how does polarization vary with temperature. What is ionic polarizability, dielectric loss and dielectric breakdown?	(4)
4B)	The polarizability of NH_3 molecule is found experimentally by the measurement of dielectric constant as 2.42×10^{-39} Fm^2 at 309K and 1.74×10^{-39} Fm^2 at 448K respectively. Calculate for each temperature the polarizability due to permanent dipole moment and due to deformation of molecules.	(4)
4C)	The Fermi energy of copper at 300 K is 7.05 eV. What is the average energy of a conduction electron in copper at 300 K?	(2)
5A)	Explain the significance of surface area to volume ratio for a nano particle using an example.	(4)
5B)	Sodium is a monovalent metal having a density of 971 kg/m3 and a molar mass of 0.023 kg/mol. Use this information to calculate the density of charge carriers and the Fermi energy.	(4)
5C)	The particle has zero energy at the lowest level according to quantum mechanics. State whether the statement is true or false and provide evidence that supports it.	(2)

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